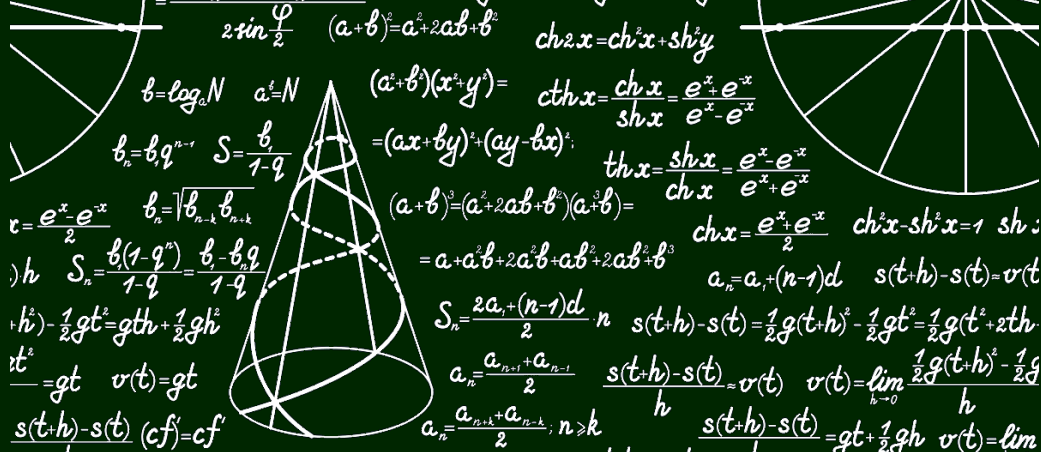




$$2r \sin \frac{\varphi}{2} \quad (a+b)^2 = a^2 + 2ab + b^2 \quad \cosh 2x = \cosh^2 x + \sinh^2 x$$

$$b = \log_a N \quad a^b = N \quad (a^2 + b^2)(x^2 + y^2) =$$

$$b_n = b_1 q^{n-1} \quad S = \frac{b_1}{1-q} \quad = (ax + by)^2 + (ay - bx)^2 \quad \cosh x = \frac{\cosh x + \sinh x}{e^x - e^{-x}}$$

$$r = \frac{e^x - e^{-x}}{2} \quad b_n = \sqrt{b_{n-k} b_{n+k}} \quad (a+b)^3 = (a^2 + 2ab + b^2)(a+b) =$$

$$h \quad S_n = \frac{b_1(1-q^n)}{1-q} = \frac{b_1 - b_n q}{1-q} \quad = a + a^2 b + 2a^2 b + ab^2 + 2ab^2 + b^3 \quad \cosh x = \frac{e^x + e^{-x}}{2} \quad \cosh^2 x - \sinh^2 x = 1 \quad \sinh :$$

$$h^2 - \frac{1}{2}gt^2 = gth + \frac{1}{2}gh^2 \quad S_n = \frac{2a_1 + (n-1)d}{2} \cdot n \quad s(t+h) - s(t) = \frac{1}{2}g(t+h)^2 - \frac{1}{2}gt^2 = \frac{1}{2}g(t^2 + 2th + h^2) - \frac{1}{2}gt^2$$

$$t^2 = gt \quad v(t) = gt \quad a_n = \frac{a_{n+1} + a_{n-1}}{2} \quad \frac{s(t+h) - s(t)}{h} \approx v(t) \quad v(t) = \lim_{h \rightarrow 0} \frac{\frac{1}{2}g(t+h)^2 - \frac{1}{2}gt^2}{h}$$

$$\frac{s(t+h) - s(t)}{h} (cf)' = cf' \quad a_n = \frac{a_{n+k} + a_{n-k}}{2}; n \geq k \quad \frac{s(t+h) - s(t)}{h} = gt + \frac{1}{2}gh \quad v(t) = \lim$$

156 RCs

$$z = r(\cos \varphi + i \sin \varphi) = \cos n\varphi + i \sin n\varphi$$

$$a + bi \quad re^{i\varphi_1} re^{i\varphi_2} = r_1 r_2 e^{i(\varphi_1 + \varphi_2)} \quad M(a, b)$$

$$a + bi \quad re^{i\varphi} \quad b = r \sin \varphi$$

$$s(t) = \frac{1}{2}gt^2$$

$$y = f(x_0) + f'(x_0)(x - x_0)$$

$$z = re^{i\varphi}$$

$$\arccos x = \frac{\pi}{2} - \arcsin x$$

1. Second-Order ODE

Definition

Equations representing the relationship between some unknown functions, the derivative of those functions and the independent variable.

Order: The order of the highest derivative of the unknown function is called the order of the differential equation.

Linear: The highest degree of the unknown function and its derivatives of any order is 1. Generally, n^{th} order differential equations have the form of $F(x, y, y', \dots, y^{(n)}) = 0$

In VV156, we only need to solve some special types of ODEs.

Step 1: Find the general solution to the corresponding homogeneous ode

$y'' + py' + qy = 0$ Solve the characteristic equation: $\lambda^2 + p\lambda + q = 0$

$$\left\{ \begin{array}{l} \text{Two different real roots } \lambda_1, \lambda_2 \quad y_h = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x} \\ \text{Two equal real roots } \lambda \quad y_h = C_1 e^{\lambda x} + C_2 x e^{\lambda x} \\ \text{Two different complex roots } \alpha \pm \beta i \quad y_h = C_1 e^{\alpha x} \sin \beta x + C_2 e^{\alpha x} \cos \beta x \end{array} \right.$$

Step 2: Find one special solution to $y'' + py' + qy = f(x)$

In VV156, we only have two types of $f(x)$:

① $f(x) = e^{\lambda x} P_m(x)$

② $f(x) = e^{\lambda x} [P_n(x) \cos(\omega x) + P_l(x) \sin(\omega x)]$

For both 1, 2, we first need to check whether $\lambda(\pm\omega i)$ is the root of the characteristic equation $\lambda^2 + p\lambda + q = 0$ or not

$$f(x) = e^{\lambda x} P_m(x):$$

1	k
λ is one of the different real roots of $\lambda^2 + p\lambda + q = 0$	1
λ is the identical real roots of $\lambda^2 + p\lambda + q = 0$	2
λ is not the real root of $\lambda^2 + p\lambda + q = 0$	0

$$f(x) = e^{\lambda x} [P_n(x)\cos(\omega x) + P_l(x)\sin(\omega x)]:$$

2	k
$\lambda \pm \omega i$ are the complex roots of $\lambda^2 + p\lambda + q = 0$	1
$\lambda \pm \omega i$ are not the complex roots of $\lambda^2 + p\lambda + q = 0$	0

Apply undetermined coefficient method:

- For 1: $y_p = x^k e^{\lambda x} \cdot Q_m(x)$
- For 2: $y_p = x^k e^{\lambda x} [Q_m(x) \cos(\omega x) + R_m(x) \sin(\omega x)] \quad (m = \text{MAX}\{n, l\})$

Q_m and R_m are another polynomial of degree m

→ Calculate y'_p and y''_p to solve Q_m and R_m

Step 3: Combine y_h and y_p together

If $f(x) = f_1(x) + \dots + f_n(x)$, and $f_i(x)$ is the form of 1, 2, we can calculate y_{pi} separately.

$$y'' + 6y' + 10y = e^{-3x} \sin x$$

Solution

① $\lambda^2 + 6\lambda + 10 = 0 \quad \lambda = -3 \pm i$

Thus, the general solution is $y_h = e^{-3x}(C_1 \cos x + C_2 \sin x)$

② Observe $e^{-3x} \sin x \rightarrow \lambda = -3, \omega = 1$

Since $-3 \pm i$ are complex roots of $\lambda^2 + 6\lambda + 10 = 0$, there is $k = 1$

So $y_p = e^{-3x}x(a \cos x + b \sin x)$

$$y_p' = e^{-3x}x[(b - 3a) \cos x + (-3b - a) \sin x] + e^{-3x}(a \cos x + b \sin x)$$

$$y_p'' = e^{-3x}x[(8a - 6b) \cos x + (6a + 8b) \sin x] + e^{-3x}[(2b - 6a) \cos x - (2a + 6b) \sin x]$$

Since $y'' + 6y' + 10y = e^{-3x} \sin x$

Check the coefficient, $a = -0.5, b = 0$

$$y_s = e^{-3x}x(-0.5 \cos x)$$

③ Combine y_h and y_p

$$y = y_h + y_p = C_1 e^{-3x} \cos x + C_2 e^{-3x} \sin x - 0.5x e^{-3x} \cos x$$

2. Integral

Type1: Unboundedness

- ① If $f(x)$ is continuous on $[a, +\infty]$, for any $t > a$, the improper integral of $f(x)$ on $[a, +\infty]$ is

$$\int_a^{+\infty} f(x)dx = \lim_{t \rightarrow +\infty} \int_a^t f(x)dx$$

- ② If $f(x)$ is continuous on $[-\infty, a]$, for any $t < a$, the improper integral of $f(x)$ on $[-\infty, a]$ is

$$\int_{-\infty}^a f(x)dx = \lim_{t \rightarrow -\infty} \int_t^a f(x)dx$$

- ③ If both $\int_a^{\infty} f(x)dx$ and $\int_{-\infty}^a f(x)dx$ are convergent

$$\int_{-\infty}^{+\infty} f(x)dx = \int_a^{+\infty} f(x)dx + \int_{-\infty}^a f(x)dx$$

Type2: Discontinuous

- ① If f is continuous on $[a, b)$ and is discontinuous at b , then the improper integral on is

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$$

- ② If f is continuous on $(a, b]$ and is discontinuous at a , then the improper integral on is

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$$

- ③ If f has a discontinuity at c ($a < c < b$), and both $\int_a^c f(x) dx$ and $\int_c^b f(x) dx$ are convergent, then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

Convergence and divergence of improper integrals

If the limits mentioned above exist (as a finite number), then the improper integral is called convergent, otherwise it's divergent.

$$\textcircled{1} \int_e^\infty \frac{1}{x(\ln x)^3} dx$$

$$\textcircled{2} \int_0^3 \frac{1}{x^2 - 6x + 5} dx$$

1.

$$\begin{aligned}\int_e^\infty \frac{1}{x(\ln x)^3} dx &= \lim_{t \rightarrow \infty} \int_e^t \frac{1}{x(\ln x)^3} dx \\&= \lim_{t \rightarrow \infty} \int_1^{\ln t} u^{-3} du \quad (u = \ln x, du = dx/x) \\&= \lim_{t \rightarrow \infty} \left[-\frac{1}{2u^2} \right]_1^{\ln t} \\&= \lim_{t \rightarrow \infty} \left[-\frac{1}{2(\ln t)^2} + \frac{1}{2} \right] \\&= 0 + \frac{1}{2} = \frac{1}{2}\end{aligned}$$

Convergent

2.

$$\begin{aligned}\int_0^3 \frac{dx}{x^2 - 6x + 5} &= \int_0^3 \frac{dx}{(x-1)(x-5)} \\ &= \int_0^1 \frac{dx}{(x-1)(x-5)} + \int_1^3 \frac{dx}{(x-1)(x-5)} \\ &= I_1 + I_2\end{aligned}$$

$$\begin{aligned}I_1 &= \frac{1}{4} \lim_{t \rightarrow 1^-} \int_0^t \left(\frac{1}{x-5} - \frac{1}{x-1} \right) dx = \lim_{t \rightarrow 1^-} \left[-\frac{1}{4} \ln |t-1| + \frac{1}{4} \ln |t-5| \right]_0^t \\ &= \lim_{t \rightarrow 1^-} \left[\left(-\frac{1}{4} \ln |t-1| + \frac{1}{4} \ln |t-5| \right) - \left(-\frac{1}{4} \ln |-1| + \frac{1}{4} \ln |-5| \right) \right]_0^t = \infty\end{aligned}$$

Since I_1 is divergent, I is divergent.

Theorem 1

Suppose $f(x)$ is continuous on $[a, \infty)$, and $f(x) \geq 0$. If

$$F(x) = \int_a^x f(t)dt$$

has an upper bound on $[a, \infty)$, then the improper integral $\int_a^\infty f(x)dx$ converges.

Theorem 2: Comparison Theorem

Suppose that f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$.

(a) If $\int_a^\infty f(x)dx$ is convergent, then $\int_a^\infty g(x)dx$ is convergent.

(b) If $\int_a^\infty g(x)dx$ is divergent, then $\int_a^\infty f(x)dx$ is divergent.

Judge whether the following improper integrals converge or not. If some of them converge, calculate their value.

$$\textcircled{1} \int_1^{\infty} \frac{1}{x^4} dx$$

$$\textcircled{2} \int_0^2 \frac{dx}{(1-x)^2}$$

$$\textcircled{3} \int_1^{\infty} \frac{1}{\sqrt{x}} dx$$

$$\textcircled{4} \int_0^1 \frac{x}{\sqrt{1-x^2}} dx$$

Solutions:

$$\textcircled{1} \int_0^{\infty} \frac{dx}{x^4} = - \left[\frac{1}{3x^3} \right] = \frac{1}{3}$$

$$\textcircled{2} \int_0^t \frac{dx}{(1-x)^2} = \left[\frac{1}{1-x} \right]_0^t = \frac{1}{1-t} - 1 \text{ Diverge}$$

$$\textcircled{3} \int_1^t \frac{1}{\sqrt{x}} dx = [2\sqrt{x}]_1^t = 2\sqrt{t} - 2 \text{ Diverge}$$

$$\textcircled{4} \int_0^1 \frac{xdx}{1-x^2} = -[\sqrt{1-x^2}]_0^1 = 1$$

Calculate the following improper integral:

$$\int_0^{\infty} \frac{1 - e^{-x^2}}{x^2} dx$$

You can directly use the conclusion that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.

(How to calculate this integral will be taught in VV255.)

Solutions:

$$\begin{aligned}\int_0^{\infty} \frac{1 - e^{-x^2}}{x^2} dx &= \int_0^{\infty} (e^{-x^2} - 1) d\left(\frac{1}{x}\right) = \left[\frac{e^{-x^2} - 1}{x} \right]_0^{\infty} - \int_0^{\infty} \frac{1}{x} d(e^{-x^2} - 1) \\ &= \left[\frac{e^{-x^2} - 1}{x} \right]_0^{\infty} + 2 \int_0^{\infty} e^{-x^2} dx\end{aligned}$$

$$\lim_{x \rightarrow \infty} \frac{e^{-x^2} - 1}{x} = 0, \lim_{x \rightarrow 0} \frac{e^{-x^2} - 1}{x} = \lim_{x \rightarrow 0} \frac{-2xe^{-x^2}}{1} = 0$$

$$2 \int_0^{\infty} e^{-x^2} dx = \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$\text{Therefore, } \int_0^{\infty} \frac{1 - 3^{-x^2}}{x^2} dx = \sqrt{\pi}$$

Theorem

$$\frac{d}{dt} \int_{\alpha(t)}^{\beta(t)} f(x) dx = f(\beta(t))\beta'(t) - f(\alpha(t))\alpha'(t)$$

How to prove it?

Use the integration to deal with the combination of limits and series.

Example

$$\textcircled{1} \lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{i}{n^2}$$

$$\textcircled{2} \lim_{n \rightarrow \infty} \sqrt{n} \left(1 + \frac{1}{n} - \sum_{i=0}^n \frac{1}{n + \sqrt{i}} \right)$$

Method

① Simply Transform into Integration. $\lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{i}{n^2} = \frac{1}{n} \sum_{i=0}^n \frac{i}{n} di = \int_0^1 x dx = \frac{1}{2}$

② Use Squeeze Theorem. $ans = \lim_{n \rightarrow \infty} \sqrt{n} \sum_{i=0}^n \frac{\sqrt{i}}{n(n + \sqrt{i})}$

$$ans \leq \lim_{n \rightarrow \infty} \sqrt{n} \sum_{i=0}^n \frac{\sqrt{i}}{n^2} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^n \sqrt{\frac{k}{n}} = \int_0^1 \sqrt{x} dx = \frac{2}{3}.$$

$$ans \geq \lim_{n \rightarrow \infty} \sqrt{n} \sum_{i=0}^n \frac{\sqrt{i}}{n(n + \sqrt{n})} = \lim_{n \rightarrow \infty} \frac{n}{n + \sqrt{n}} \frac{1}{n} \sum_{i=0}^n \sqrt{\frac{k}{n}} = \frac{2}{3}.$$

3. Applications of Integration

Types of Applications to encounter in VV156

- ① Area Between Curves
- ② Volumes
- ③ Volumes by Cylindrical Shells
- ④ Average Value of a function
- ⑤ Arc Length
- ⑥ Area of a Surface of Revolution

Equation

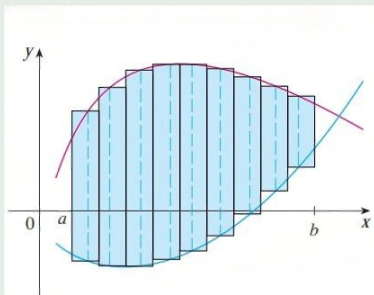
- ① The area of the region bounded by the curves $y = f(x)$, $y = g(x)$, and the lines $x = a$, $x = b$, where f & g are continuous & $f(x) \geq g(x) \quad \forall x \in [a, b]$, is

$$A = \int_a^b (f(x) - g(x)) dx$$

- ② The unsigned area between the curves $y = f(x)$, $y = g(x)$, and between $x = a$, $x = b$ is

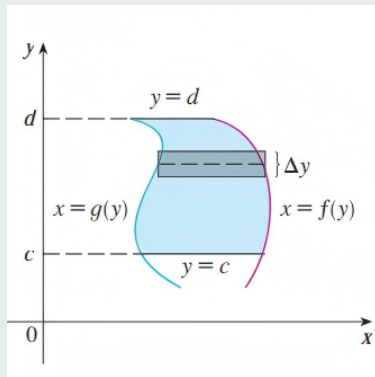
$$A = \int_a^b |f(x) - g(x)| dx$$

Some regions are best treated by regarding x as a function of y .



(b) Approximating rectangles

$$A = \int_a^b |f(x) - g(x)| dx$$



$$A = \int_c^d |f(y) - g(y)| dy$$

Find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

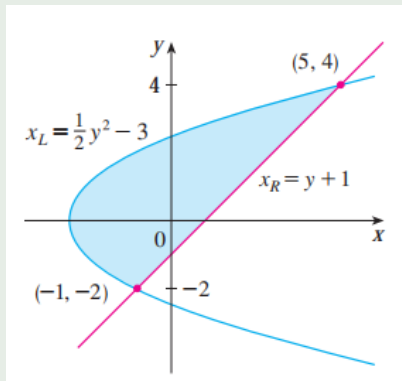
Method 1: Integrate horizontally

The left and right boundary curves:

$$x_L = \frac{1}{2}y^2 - 3 \quad \text{and} \quad x_R = y + 1$$

Integrate between the appropriate y -values, $y = -2$ and $y = 4$:

$$\begin{aligned} A &= \int_{-2}^4 (x_R - x_L) dy \\ &= \int_{-2}^4 \left[(y + 1) - \left(\frac{1}{2}y^2 - 3 \right) \right] dy = 18 \end{aligned}$$



Method 2: Integrate vertically

The left upper and lower boundary curves:

$$y_{U_1} = \sqrt{2x+6} \quad \text{and} \quad y_{L_1} = -\sqrt{2x+6}$$

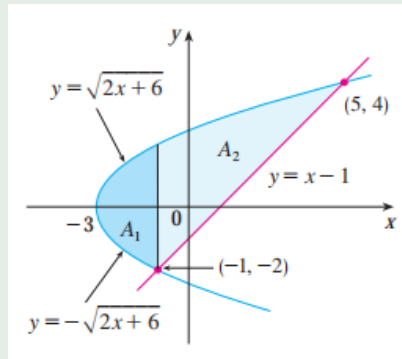
The right upper and lower boundary curves:

$$y_{U_2} = \sqrt{2x+6} \quad \text{and} \quad y_{L_2} = x-1$$

Integrate between the appropriate sections: $[-3, -1]$
and $[-1, 5]$

$$A_1 = \int_{-3}^{-1} (y_{U_1} - y_{L_1}) dx \quad A_2 = \int_{-1}^5 (y_{U_2} - y_{L_2}) dx$$

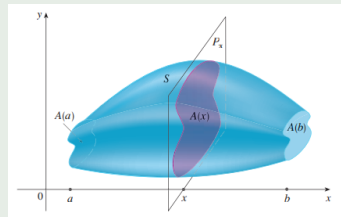
Comment: Very complicated, so choose wisely at first!



Equations

Let S be a solid that lies between $x=a$ and $x=b$. If the cross-sectional area of S in the plane, through x and perpendicular to the x -axis is $A(x)$, where A is a continuous function, then the volume of S is

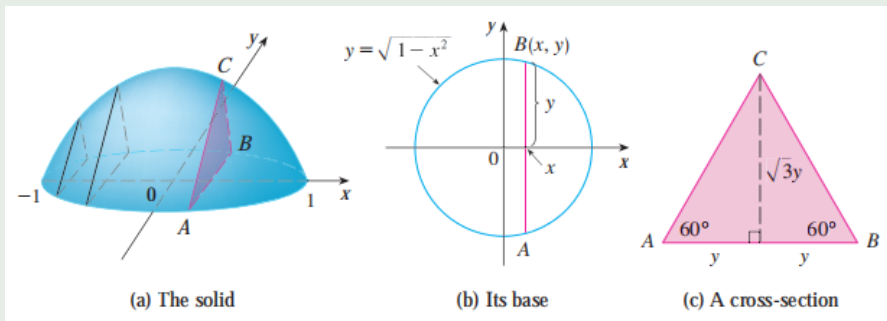
$$V = \int_a^b A(x) dx$$



Imagine cutting the block in pieces, then integrate the area of those pieces.

Consider a solid with a circular base of radius 1. Parallel cross-sections perpendicular to the base are equilateral triangles. Find the volume of the solid.

Step 1: Imagine the solid and draw the figure.



Step 2: Integrate

$$A(x) = \frac{1}{2} \cdot 2\sqrt{1-x^2} \cdot \sqrt{3}\sqrt{1-x^2} = \sqrt{3}(1-x^2)$$

$$V = \int_{-1}^1 A(x) dx = \int_{-1}^1 \sqrt{3}(1-x^2) dx = \frac{4\sqrt{3}}{3}$$

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